Aditya Chaturvedi

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Profile

Master's student in Mechatronics and Cyber-Physical Systems with a strong interest in XR technologies, 3D simulation, and digital engineering tools. Experienced in developing VR-based training applications using Unity and Blender and working with CAD models in immersive environments. Skilled in 3D modelling, simulation, and robotics systems, with programming experience in Python and ROS. Highly motivated to contribute to XR-based industrial software development and process simulation research.

Education

Deggendorf Institute of Technology — Deggendorf, Germany

M. Eng. Mechatronics and Cyber-Physical Systems — Grade-1.8

Technical Skills

XR & 3D Development: Unity (VR development, interactive simulation), Blender (3D modelling, asset optimization), VR application development, 3D CAD integration into virtual environments
Programming & Scripting: Python, MATLAB, Simulink, Linux (basic)
Robotics & Automation: ROS (Robot Operating System), Gazebo Simulation, URDF Modelling, Autonomous Navigation, Sensor Integration (Camera, LiDAR)
Mechanical Design & Simulation: SolidWorks, CATIA, ANSYS (FEA), System Modelling, Topology Optimisation, Additive Manufacturing (FDM)
Core Competencies: Mechatronic System Design, Robotics Integration, Process Optimisation, Problem Solving, Continuous Improvement.

Professional Experience

Production Supervisor — Jul 2023 – Feb 2024

Rajshree Engineering Pvt. Ltd., Pithampur, India

- Orchestrated production planning and ensured strict schedule adherence to meet daily and weekly manufacturing targets.
- Executed daily defect analysis and applied immediate countermeasures to resolve technical bottlenecks and minimise downtime.
- Led continuous improvement initiatives by generating weekly "Top 5 Defect Analysis" reports, utilising Kaizen and PDCA (Plan-Do-Check-Act) frameworks to enhance productivity and quality.
- Optimised resource allocation using the 4M methodology (Man, Machine, Material, Method) to achieve safety compliance and operational KPIs.

Graduate Engineer Trainee — Mar 2022 – May 2023

TATA Motors, Pune, India

- Collaborated in the design validation and functional testing of new automotive components and systems to ensure compliance with engineering specifications.
- Implemented Lean Manufacturing principles to identify process bottlenecks and significantly reduce operational waste on the production line.
- Authored detailed technical documentation, including daily production reports, defect analysis logs, and project status updates to support cross-functional teams.

Graduate Engineer Trainee — Mar 2021 – Sep 2021

Wonder Cement, Rajasthan, India

- Acquired hands-on expertise in the cement manufacturing lifecycle through rigorous training and on-site operational involvement.
- Supported production planning strategies and monitored daily production operations to ensure workflow continuity and efficiency.
- Managed technical documentation and generated comprehensive operational reports to assist management in process tracking.

Academic Projects

VR Safety Training for 3D Printer Operations — **Oct** 2024 - Jan 2025

Deggendorf Institute of Technology, Cham

- Developed an immersive VR training application using Unity to simulate safe operation procedures for industrial Stratasys 3D printers.
- Created and optimized high-fidelity 3D assets using Blender and SolidWorks, ensuring real-time performance in a VR environment.
- Implemented interactive training scenarios and hazard simulations to enhance operator learning and safety awareness.
- Tools: SolidWorks, Blender, Unity.

Dryer Door Push Latch Mechanism **Oct** 2024 - Jan 2025

Deggendorf Institute of Technology, Cham

- Designed and topology-optimised a push-lock/push-release dryer latch mechanism using SolidWorks, engineered to withstand high operating temperatures and a lifecycle of 28,000 cycles.
- Prototyped the component via Additive Manufacturing (FDM) using ABS material, optimising the design for weight reduction, ease of assembly, and cost-efficient manufacturing.
- Tools: SolidWorks, Additive Manufacturing (FDM), Topology Optimisation, Ultimaker Cura

Case Study Cooperative and Autonomous Systems Mar 2024 – Jul 2024

Deggendorf Institute of Technology, Cham

- Designed and modeled a differential-drive robot in SolidWorks and converted it into a URDF simulation model.
- Implemented robot simulation and navigation algorithms using ROS and Gazebo.
- Integrated LiDAR and camera sensors for environmental perception and obstacle avoidance.
- Conducted simulation experiments to evaluate robot mobility and autonomous navigation performance
- Tools: SolidWorks, ROS, Gazebo, URDF

Case study Mechatronic System Simulation Mar 2024 – Jul 2024

Deggendorf Institute of Technology, Cham

- Simulation of the soil's moisture content by comparing bare soil and mulched soil.
- Approx. 20% more moisture retention in mulched soil in comparison to bare soil.
- SolidWorks, Gazebo, ROS

Design Analysis and Fabrication of a Single Piston Floating Calliper of the Hydraulic Brake System of an All-Terrain Vehicle 2018 –2019

KIET Group of Institutes, India

- To get the required amount of brake force for our customised vehicle.
- Modular Al 7075-T6 calliper design reduced thermal deformation, machining cost, and thermal displacement by 8.56% vs. conventional.
- SolidWorks, Ansys

Achievements & Honors

- Secured All India Rank 07 and 1st position in CAE and Sled Pull at SAEINDIA BAJA 2018.
- Secured All India Rank 24 and 3rd position in the Cost Event at SAEINDIA BAJA 2017.
- Active SAEINDIA member (2015–2018) with participation in national-level engineering competitions and technical events.
- 3rd Place – CAD Competition (Autodesk Inventor)

Interests

- Focus Areas: Robotics and autonomous systems, mechatronic system design, intelligent automation, simulation and modeling, additive manufacturing
- Continuous Learning: Active learner in robotics, cybersecurity fundamentals, and automation technologies through academic projects, case studies, and self-directed technical exploration
- Multilingual: Professional English proficiency, German B1 (intermediate), experienced in cross-cultural academic and industrial collaboration.
- XR Technologies (AR/VR), 3D simulation and digital twins, Industrial VR training systems, Metaverse applications in manufacturing, Robotics simulation